

Total no. of Pages:

Roll no.

MID Semester of 3rd Sem B.Tech.

MID –SEMESTER EXAMINATION, SEPTEMBER, 2024

COURSE CODE: EPMT301

COURSE TITLE: MATHEMATICS FOR MACHINE LEARNING

TIME: 1.30 HRS

MAX. MARK: 20

Note:- Attempt all questions. Missing data/ information (if any), may be suitably assumed & mentioned in the answer.

Q. NO	QUESTION	MARKS	CO
1a	$T: R^2 \rightarrow R^2$ defined by $T(x, y) = (x + y, x - y)$. Determine the matrix of the linear transformation T with respect to the standard ordered basis of R^2 .	2	CO1
1b	Examine whether the subset $W = \{(x, y, z, w) \in R^4: x + y + z + w = 1\}$ is a subspace of R^4 .	2	CO1
2a	Determine the value of k such that the subset $S = \{(1, 2, 1), (k, 3, 1), (2, k, 0)\}$ of R^3 is linearly dependent.	2	CO1
2b	Find Kernel of the linear mapping $T: R^3 \rightarrow R$ defined by $T(x, y, z) = (x + y + z)$.	2	CO1
3a	Examine whether T is a linear transformation/ mapping where $T: R^2 \rightarrow R^2$ is defined by $T(x, y) = (y, 1)$.	2	CO1
3b	If $\langle \alpha, \beta \rangle = 0$, prove that $\ \alpha + \beta\ ^2 = \ \alpha\ ^2 + \ \beta\ ^2$ where vectors α, β are in real vector space with real inner product.	2	CO2
4a	Prove that the quadratic form $Q(x, y, z) = x^2 + y^2 + 2xy + 2xz + 2yz$ is neither positive definite nor positive semi-definite.	2	CO2
4b	Find eigen values of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$	2	CO2
5a	Using Gram-Schmidt process, convert the basis $\{(1, 2 - 2), (2, 0, 1), (1, 1, 0)\}$ of the Euclidean space R^3 with the standard inner product into an orthogonal basis $\{w_1, w_2, w_3\}$ where $w_1 = (1, 2 - 2)$.	2	CO2
5b	Find the value of a and b such that the set $S = \{(1, 1, a, 1), (-2, -1, 1, 2), (b, -17, -13, 14)\}$ is an orthogonal subset of R^4	2	CO2